silicon, typically about 1 cm square. As shown by the cross-sectional view in Fig. 23-7, the backside is coated with a thin layer of metal connected to ground potential. The topside is covered with a thin electrical insulator, and a repetitive pattern of electrodes. The most common type of CCD is the **three phase readout**, where every third electrode is connected together. The silicon used is called *p-type*, meaning it has an excess of positive charge carriers called *holes*. For this discussion, a hole can be thought of as a positively charged particle that is free to move around in the silicon. Holes are represented in this figure by the "+" symbol.

In (a), +10 volts is applied to one of the three phases, while the other two are held at 0 volts. This causes the holes to move away from every third electrode, since positive charges are repelled by a positive voltage. This forms a region under these electrodes called a **well**, a shortened version of the physics term: *potential well*.

Each well in the CCD is a very efficient light sensor. As shown in (b), a single photon of light striking the silicon converts its energy into the formation of two charged particles, one electron, and one hole. The hole moves away, leaving the electron stuck in the well, held by the positive voltage on the electrode. Electrons in this illustration are represented by the "-" symbol. During the **integration period**, the pattern of light striking the CCD is transferred into a pattern of charge within the CCD wells. Dimmer light sources require longer integration periods. For example, the integration period for standard television is 1/60th of a second, while astrophotography can accumulate light for many hours.

Readout of the electronic image is quite clever; the accumulated electrons in each well are *pushed* to the output amplifier. As shown in (c), a positive voltage is placed on *two* of the phase lines. This results in each well expanding to the right. As shown in (d), the next step is to remove the voltage from the first phase, causing the original wells to collapse. This leaves the accumulated electrons in one well to the right of where they started. By repeating this pulsing sequence among the three phase lines, the accumulated electrons are pushed to the right until they reach a **charge sensitive amplifier**. This is a fancy name for a capacitor followed by a unity gain buffer. As the electrons are pushed from the last well, they flow onto the capacitor where they produce a voltage. To achieve high sensitivity, the capacitors are made extremely small, usually less than 1 ρF. This capacitor and amplifier are an integral part of the CCD, and are made on the same piece of silicon. The signal leaving the CCD is a sequence of voltage levels proportional to the amount of light that has fallen on sequential wells.

Figure 23-8 shows how the two-dimensional image is read from the CCD. After the integration period, the charge accumulated in each well is moved up the column, one row at a time. For example, all the wells in row 15 are first moved into row 14, then row 13, then row 12, etc. Each time the rows are moved up, all the wells in row number 1 are transferred into the **horizontal register**. This is a group of specialized CCD wells that rapidly move the charge in a horizontal direction to the charge sensitive amplifier.